What Should Science Ask Next?

Prashant Garg
Imperial College London
github.com/prashgarg/frontiergraph

Economics seminar version

- question choice is under-formalized
- the literature is large and fragmented
- some progress adds mechanisms, some states direct links
- the graph helps rank what may be worth reading first

Appendix

Question choice is a science-wide bottleneck

- Choosing what to work on is one of the least formalized decisions in science.
- The burden of existing knowledge keeps rising as the stock of work grows.
- Conservative search dominates even when riskier exploration matters disproportionately.
- If AI lowers downstream costs, the scarce input shifts upstream toward question choice.

Three anchor ideas

why this is not just an economics problem

- knowledge burden rises
- ideas get harder to find
- scientists still search conservatively

Jones (2009); Bloom et al. (2020); Foster, Rzhetsky, and Evans (2015)

Appendix

Why this setting?

- The paper uses a published economics-facing corpus because it gives dated, realized research structure.
- That makes missing-question backtesting possible without mixing in draft noise.
- The object is economics-facing, but the broader question is science-wide.

Why published papers?

what this setting buys

- dated structure
- observed realizations
- clean prospective evaluation

Narrow enough to interpret, broad enough to matter.

Appendix

This paper

We build

data object

- a dated graph from published economics-facing papers
- paper-local extraction, then cross-paper matching

We test

historical screen

- rank still-open next questions using only the literature observed at date t-1
- evaluate them prospectively against later publication

We learn

substantive implication

- the graph helps in two distinct historical families
- the historical result reveals which kind of scientific movement is more common

The seminar centers the substantive interpretation, not only the benchmark.

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Appendix

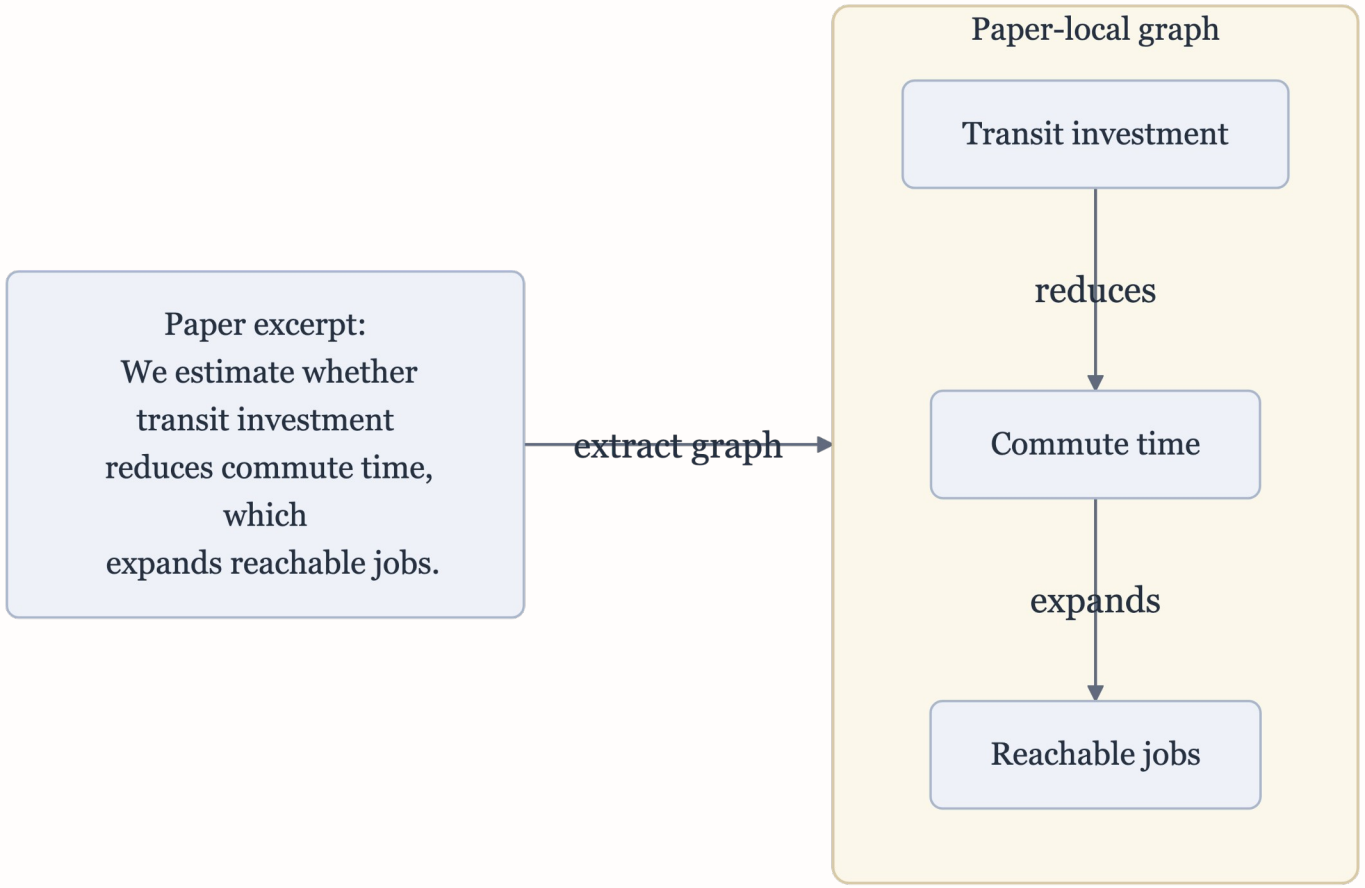
Overall pipeline

- Extraction builds one local graph per paper.
- Matching creates a shared concept space across papers.
- Candidate generation freezes the graph at t-1 and ranks still-open directed questions.

This is the full-paper object in one slide, before we split into the two historical families.

Appendix

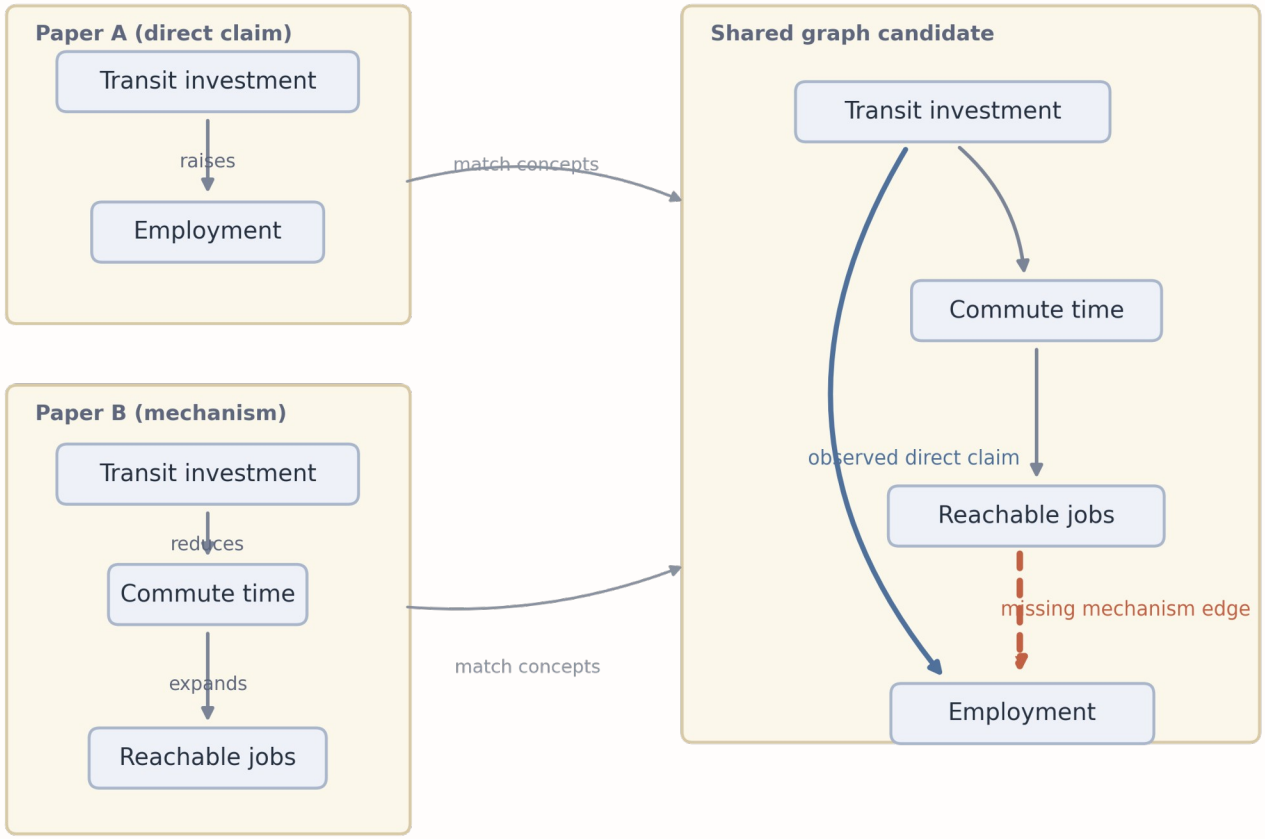
Each paper becomes a local graph



Paper-local extraction gives the first reusable object.

Appendix

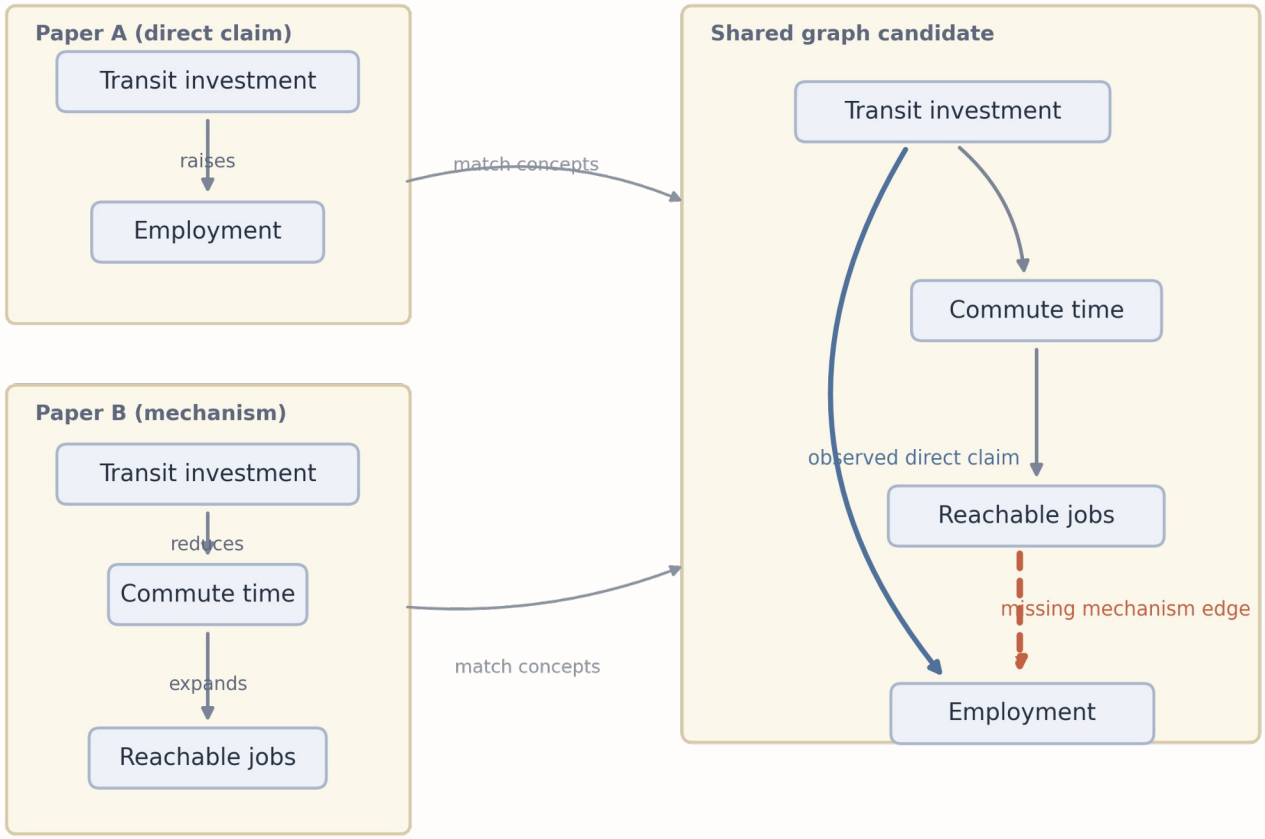
Candidate questions appear only after cross-paper matching



Cross-paper matching is what turns paper-local fragments into shared candidate structure.

Appendix

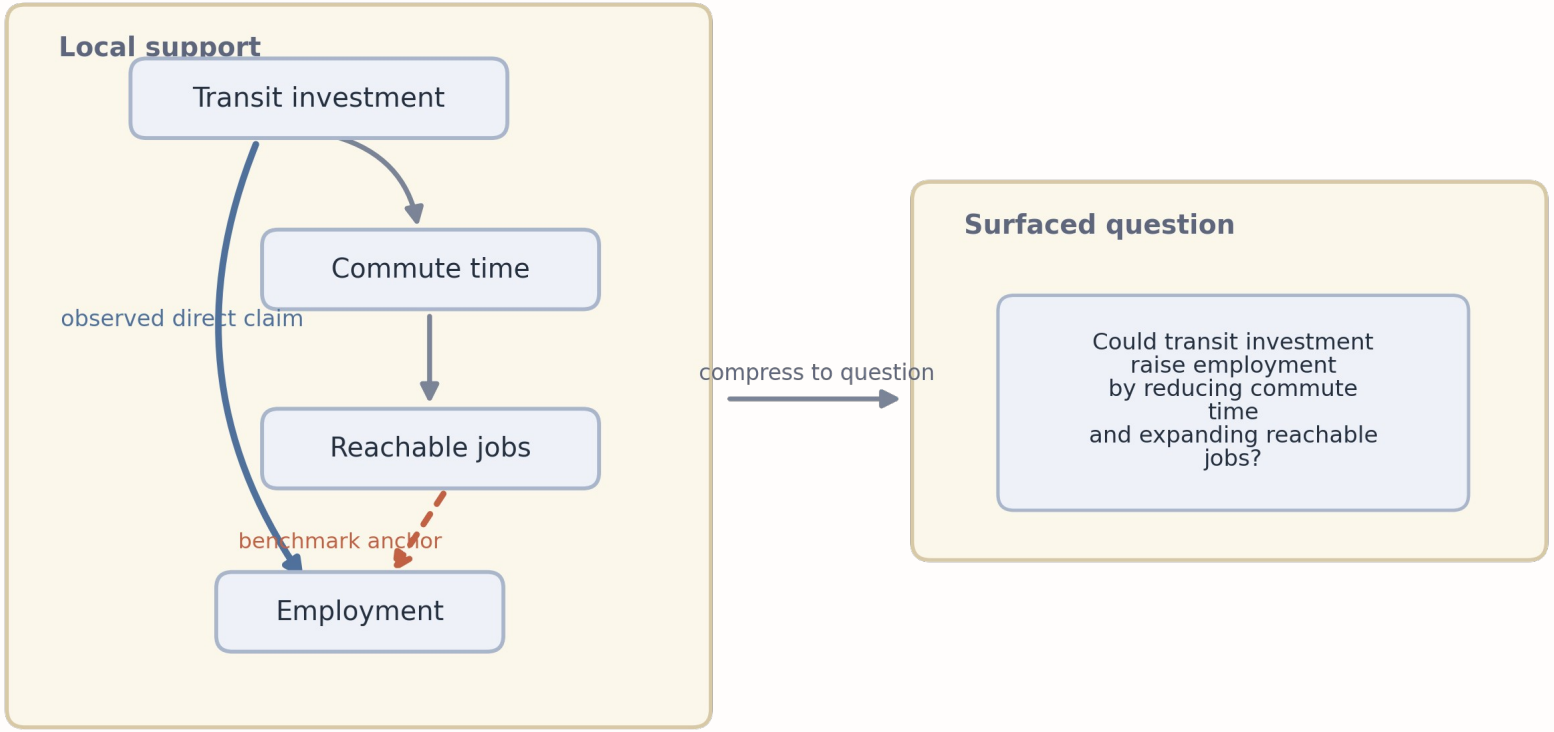
Why node normalization matters



Normalization is central because candidate generation, paths, and missingness all depend on concept identity.

Empirical object

Two graph-grounded next-question objects



The benchmark event is narrower than the surfaced question.

Appendix

Historical evaluation design

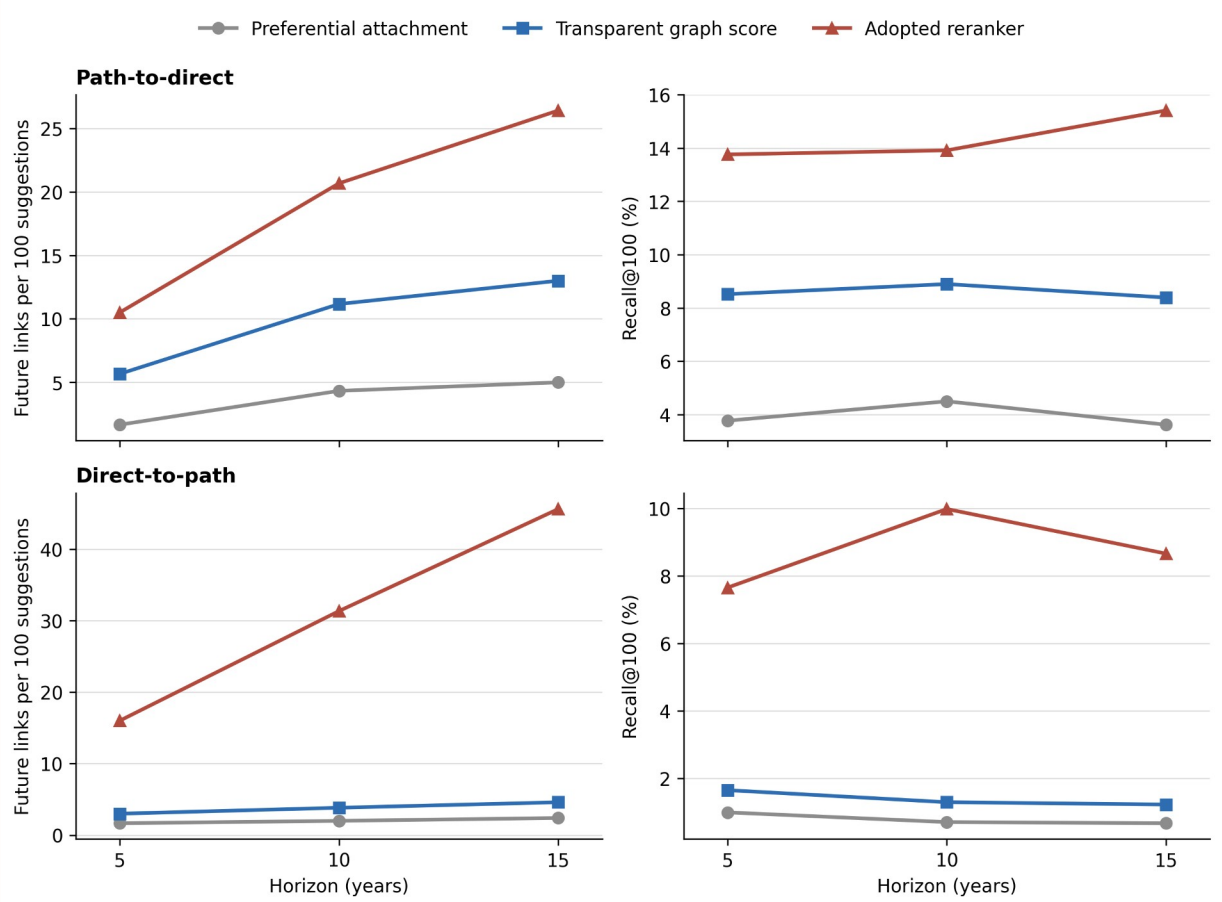
- Freeze the graph at t-1.
- Rank still-open directed candidates.
- Ask which questions later appear during [t, t+h].
- Keep the benchmark event separate from the richer reader-facing surfaced question.

This is what makes the exercise prospective rather than narrative.

Historical result

Appendix

Paired historical benchmark

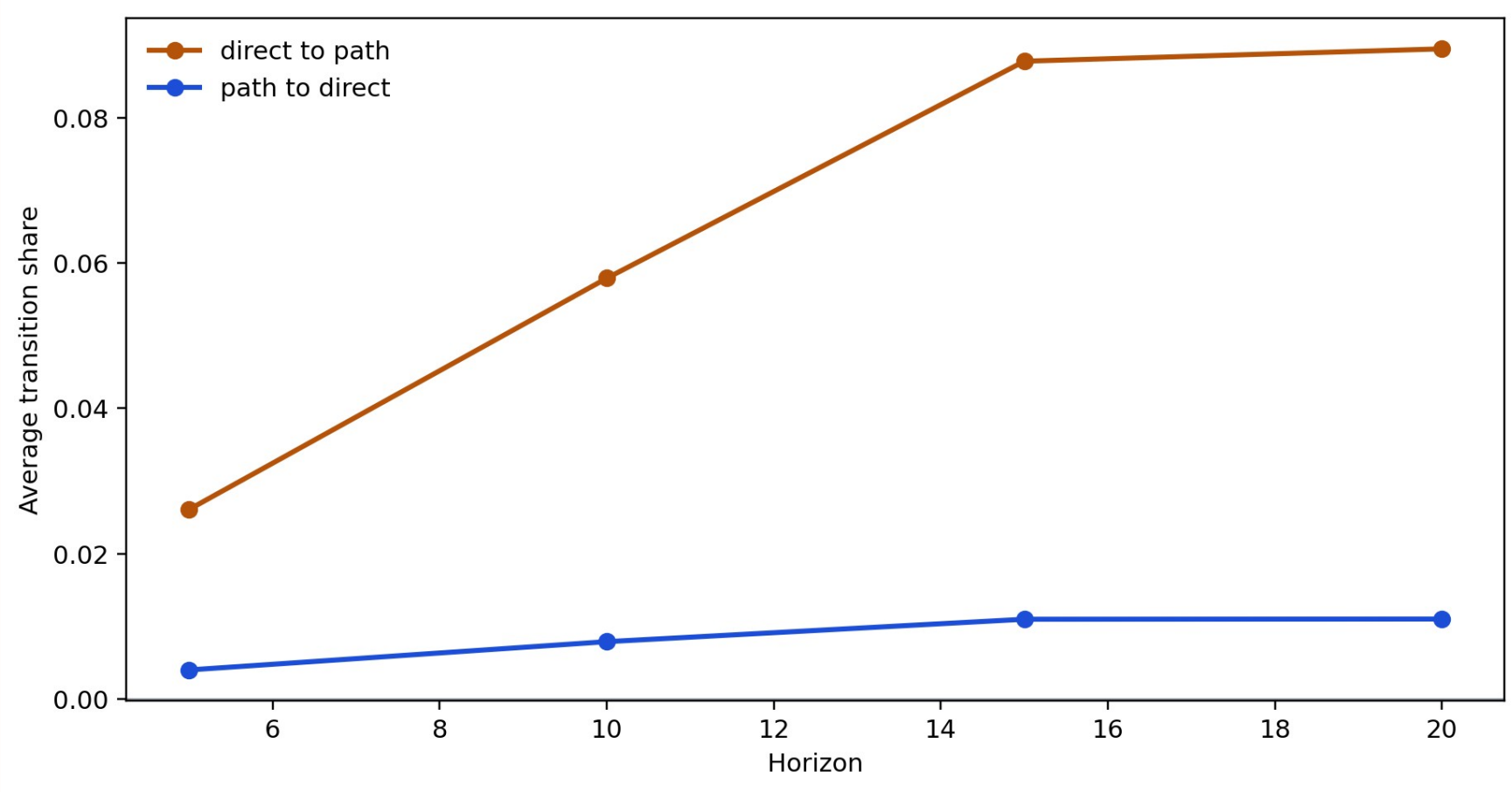


Graph-based screening improves on popularity in both families.

Substantive result

Appendix

How the literature moves

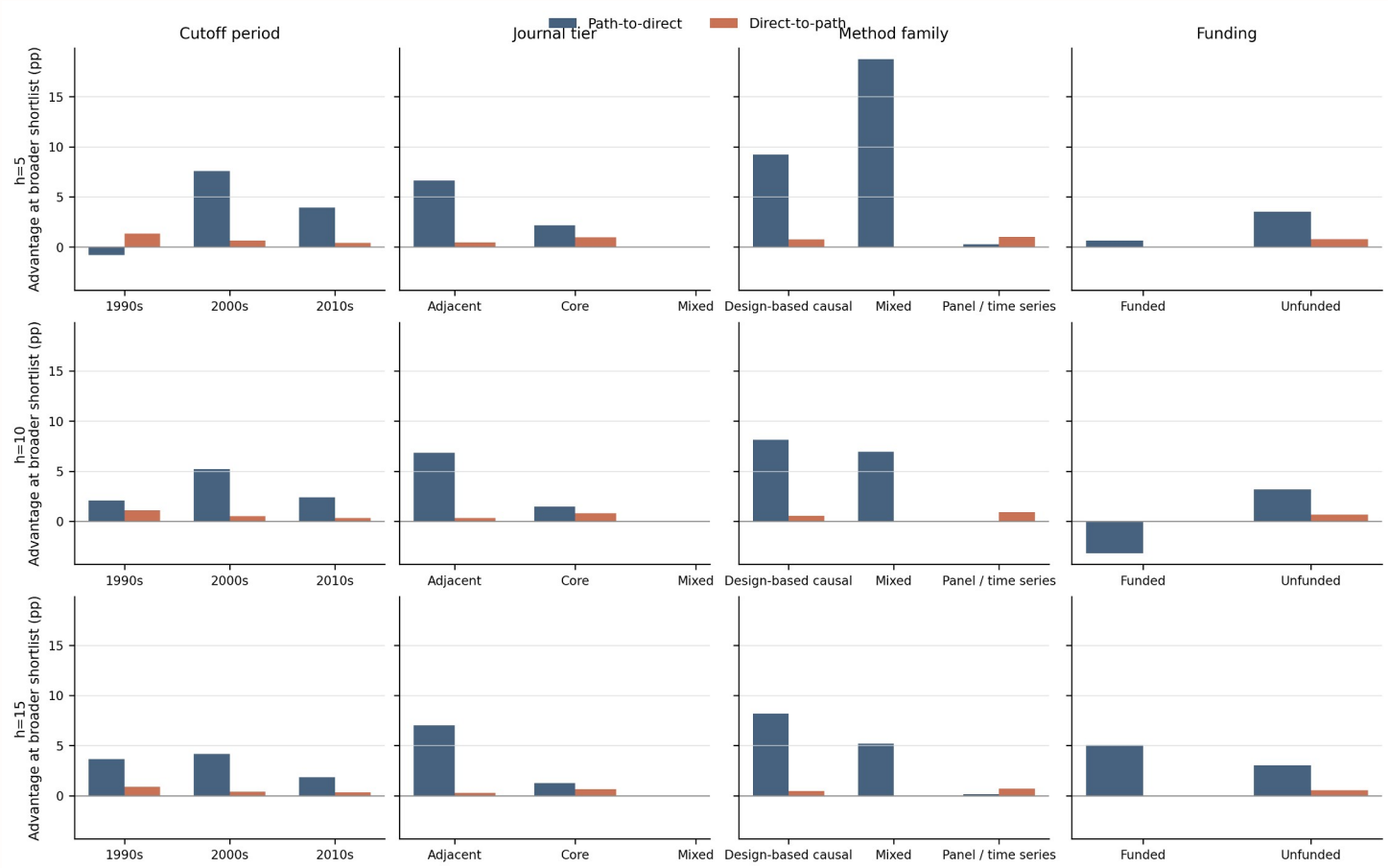


The literature more often thickens mechanisms than it later states locally implied direct links.

Extra empirical view

Appendix

Where the graph helps



This stays secondary to the main historical and substantive comparison.

Appendix

Why the families differ

Direct-to-path

- smaller candidate universe
- higher later realizations per shortlist
- benchmark event: a mechanism appears around an already known relation

Path-to-direct

- larger candidate universe
- higher recall of all later-realized links
- benchmark event: a locally supported direct relation later becomes explicit

These are different denominators, not inconsistent results. One family is better for finding high-hit mechanism thickening; the other is better for recovering a broader set of later explicit links.

Direct-to-path and path-to-direct are not mirror images; they reward different kinds of success.

Appendix

What the graph surfaces now

Illustrative cleaned renderings of current frontier items

Direct-to-path

known relation, missing mechanism

- Could digital financial inclusion support employment through labour matching or market access?
- Could tax refunds lift productivity through liquidity and investment?
- Could nutrition information shift willingness to pay through health-salience channels?

Path-to-direct

nearby support, missing direct relation

- Could trade liberalization raise R&D?
- Could the digital economy strengthen environmental regulation?
- Could renewable energy reshape urbanization?

The path-to-direct side is already cleaner publicly. The direct-to-path side is promising but still needs more manual relabeling.

Appendix

What this changes

- Question choice can be partly disciplined rather than treated as pure intuition.
- The graph is most useful for directing scarce reading time, not replacing judgment.
- Mechanism thickening appears to be a central mode of scientific progress.

Practical reading rule

how I would actually use this

- use the graph to choose what to inspect first
- compare the two families rather than forcing one object
- treat ranked questions as a disciplined shortlist, not an oracle

This is the slide to linger on in discussion.

Appendix

What this does not do

Not all imagination

It does not rank all scientific imagination.

Imperfect identity

It does not solve concept identity perfectly.

Judgment still matters

It does not remove the need for field knowledge or judgment.

Cleanly dateable only

It benchmarks only objects that can be dated cleanly in the observed literature.

The limits are part of the design, not afterthoughts.

Appendix

Where this can go next

Longer path support

ask whether slightly more distant support improves novelty without losing too much coherence

Context extension

take known relations and ask where they remain under-tested across settings

LLM workflow layers

use screening models as a practical refinement layer on top of graph-ranked shortlists

Node activation

treat the appearance of entirely new concepts as a separate frontier object

These are research directions, not slide filler.

Appendix

Conclusion

Build

Build a dated graph from published research.

Test

Use it to rank still-open next questions prospectively.

Learn

Learn not only whether the graph works, but what kind of scientific movement is more common.

The benchmark is the credibility layer. The contribution is the claim about how science moves and how researchers can choose what to inspect next.

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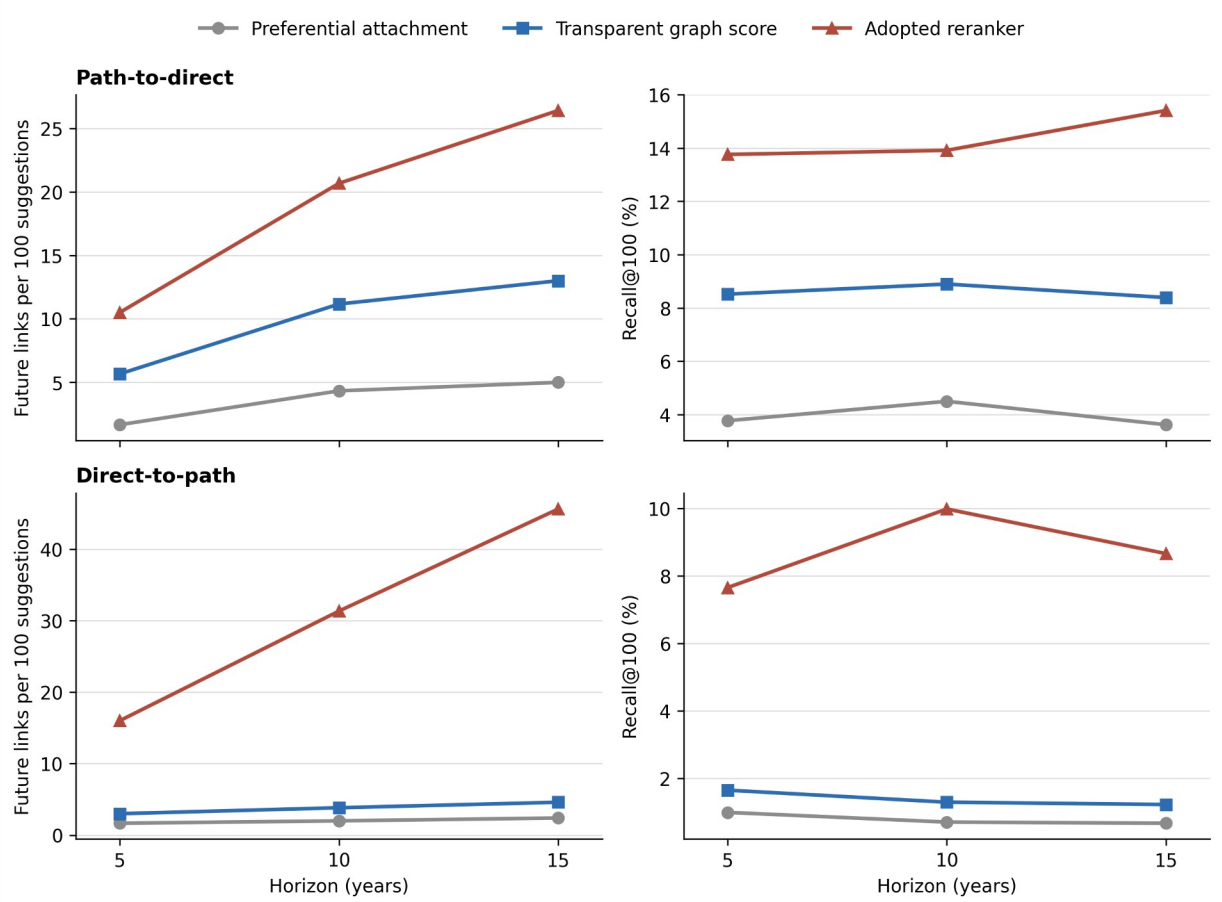
Extra benchmark detail



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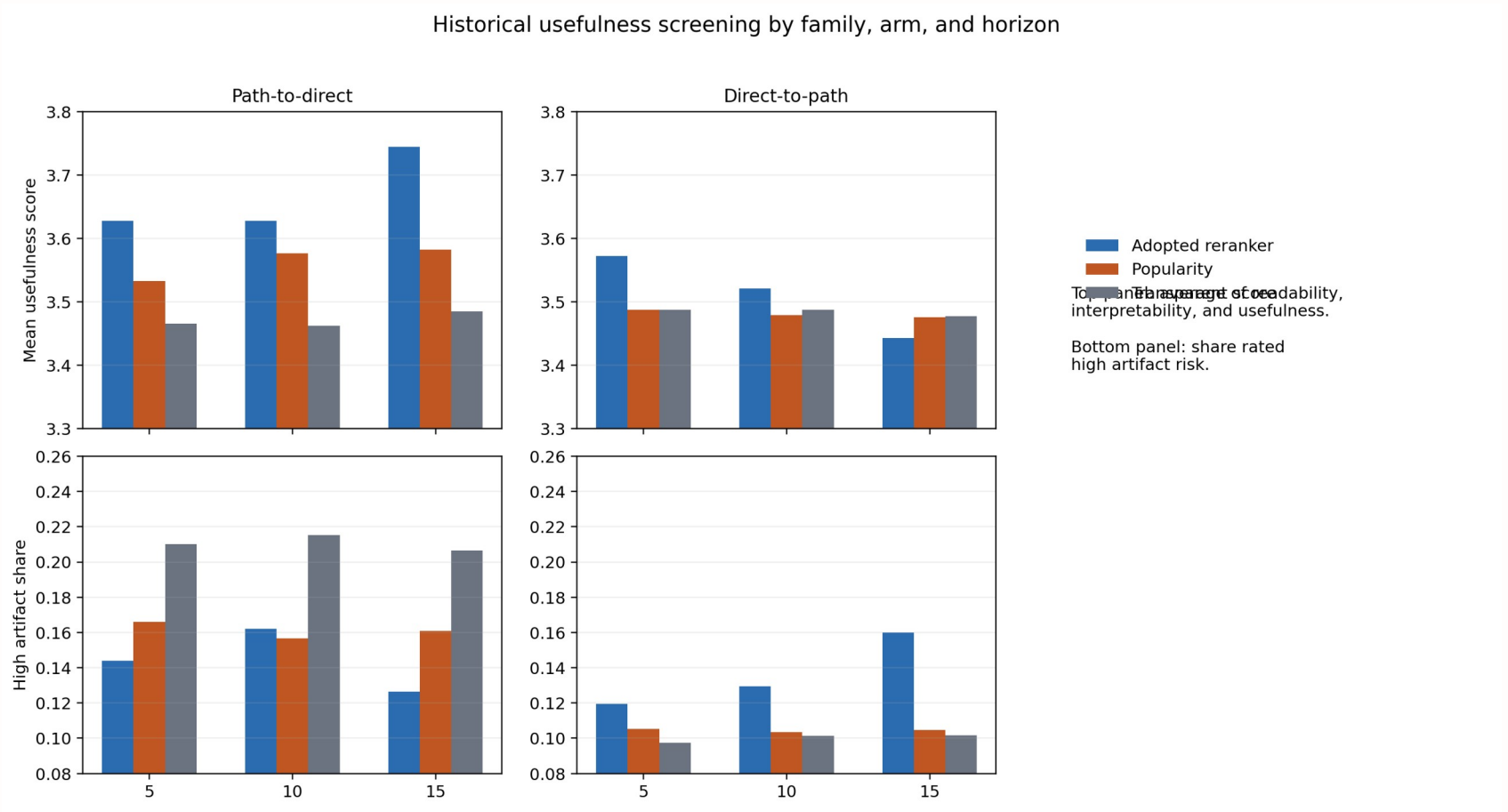
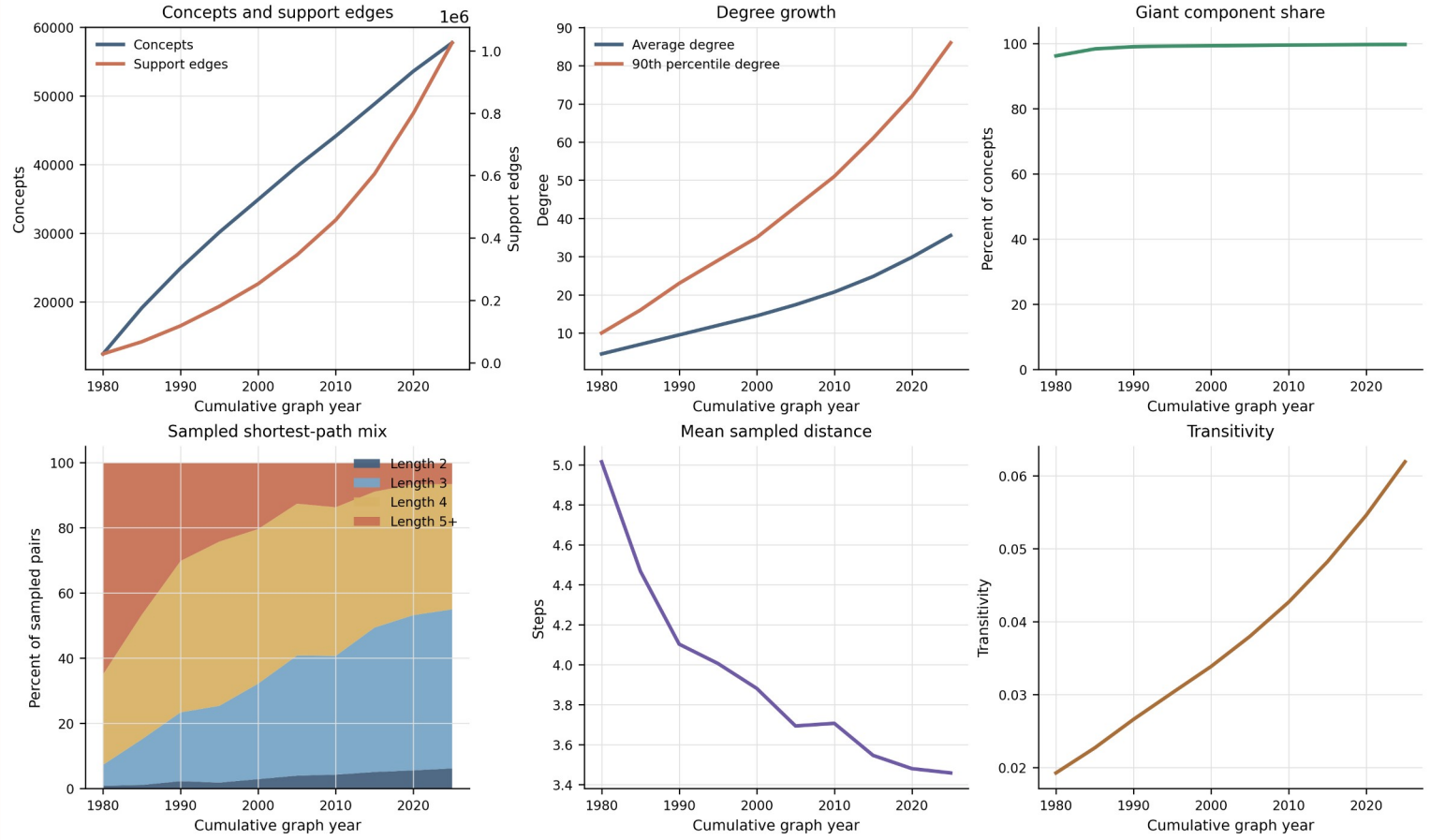
Current frontier and usefulness



Graph evolution

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